CS5234 Quiz 2 · Cheat Sheet

Lectures 5, 7–10 · Nov 5, LT12 · open book · log = ln

0 · Recognise the problem → tool

- **Streaming upper bound**: keep a sketch / sample / summary; unbiased estimator → variance → Chebyshev (mean trick) → median trick
- **Streaming lower bound**: one-way communication; reduce from Index / Gap-Hamming / Disjointness; encode Bob's index as a key edge / element
- **Deletions allowed (turnstile)**: use a linear sketch Rx
- **Counting with tiny memory**: Morris: store log of the count; mean trick
- **Unknown scale**: convert to a testing problem with threshold T, run $T = (1+\varepsilon)^j$ in parallel
- **Graph stream**: keep a sparse subgraph (forest / spanner) and prove an invariant + an edge-count bound
- **How much independence?** Chebyshev only → pairwise; 4th moment (AMS) → 4-wise; conditioning on k heavy items → $(k+2)$-wise

1 · Toolbox recap

- Markov $\Pr(X \geq t) \leq \mathbb{E}X/t$; Chebyshev $\Pr(|X-\mathbb{E}X| \geq t) \leq \mathrm{Var}X/t^2$
- Mean of k i.i.d.: Var/k; need $k = \frac{3\mathrm{Var}X}{\varepsilon^2(\mathbb{E}X)^2}$ for 2/3
- Chernoff (indep., $[0,1]$): $\Pr(|X-\mu| \geq \delta\mu) \leq 2e^{-\delta^2\mu/3}$
- Median trick: success $> \frac{1}{2}$ → $O(\log\frac{1}{\delta})$ copies; union bound over m items with $\delta = 1/m^2$
- $\{0,1\}$ var.: Var $\leq \mathbb{E}$; pairwise indep. ⇒ variances add
- $(1-\frac{1}{T})^T \approx \frac{1}{e}$, $(1-\frac{1}{T})^{2T} \leq \frac{1}{e^2}$; $\frac{d^k}{k!} \leq (\frac{ed}{k})^k$

2 · L5 Streaming basics

Model elements arrive one by one, seen once; memory ideally polylog(n); storing one element $\log m$ bits, a counter $\log n$ bits.

Reservoir sampling keep x; when x_i arrives replace with prob. $\frac{1}{i}$.
$\Pr[\text{final} = x_i] = \frac{1}{i} \cdot \frac{i}{i+1} \cdots \frac{n-1}{n} = \frac{1}{n}$. Space $\log m + \log n$. k independent copies ⇒ k samples with replacement.

Morris counter $C \leftarrow C+1$ w.p. 2^{-C} on each 1; output $2^C - 1$.
$X_i = 2^{C_i}$: $\mathbb{E}[X_{i+1} \mid X_i{=}x] = 2x \cdot \frac{1}{x} + x(1-\frac{1}{x}) = x+1 \Rightarrow \mathbb{E}X_A = A+1$ (unbiased).
$\mathbb{E}[X_{i+1}^2 \mid x] = x^2 + 3x \Rightarrow \mathbb{E}X_i^2 = \frac{3i(i+1)}{2} + 1$, $\mathrm{Var}X_A = \frac{A(A-1)}{2} < A^2$.
Mean of $\frac{10}{\varepsilon^2}$ copies: Var $< \frac{\varepsilon^2 A^2}{10}$ ⇒ Chebyshev $\Pr[|\bar{A}-A| > \varepsilon A] < \frac{1}{10}$. Space $O(\varepsilon^{-2}\log\log n)$ ($C \leq \log n$ w.h.p.). **Key step: law of total expectation, then induction.**

Edge stream vertex set known, edges arrive one by one.

Connectivity keep spanning forest F: add e iff no cycle. $\leq n-1$ edges ⇒ $O(n\log n)$ bits; G connected ⇔ F connected.

Bipartiteness bipartite ⇔ no odd cycle. Keep forest F; if $F \cup e$ has an odd cycle → "No"; no cycle → add. Correctness: "Yes" ⇒ 2-colour the forest; every $e \notin F$ closes an even cycle ⇒ endpoints differ ⇒ colouring valid for G.

$(2k-1)$-spanner $H \subseteq G$ with $d_G \leq d_H \leq (2k-1)d_G$. Add $e=(u,v)$ iff it closes no cycle of length $\leq 2k$ (i.e. $d_H(u,v) > 2k-1$).
Stretch: a rejected edge already has $d_H(u,v) \leq 2k-1$; replace each edge of a shortest path ⇒ $d_H \leq (2k-1)d_G$.
Size: girth $> 2k$ ⇒ $O(n^{1+1/k})$ edges. (i) min degree δ: first k BFS layers are a tree with $\geq (\delta-1)^k \leq n$ vertices ⇒ $\delta = O(n^{1/k})$. (ii) average degree d ⇒ peel vertices of degree $< \frac{d}{4}$ (removes $< \frac{m}{2}$ edges) ⇒ subgraph with min degree $\geq \frac{d}{4}$ ⇒ $d = O(n^{1/k})$.
Space $O(n^{1+1/k}\log n)$ bits. $k=2$: 3-spanner, $O(n^{3/2})$ edges.

3 · L7 Distinct elements

Testing $h : [m] \to [T]$; $\Pr[\text{no element hashes to } 1] = (1-\frac{1}{T})^{DE}$: $\approx \frac{1}{e}$ if $DE \leq T$, $\leq \frac{1}{e^2}$ if $DE \geq 2T$. $k = O(1)$ copies, threshold $\frac{1}{2}(\frac{1}{e}+\frac{1}{e^2})$, Chernoff. Guess $T = (1+\varepsilon)^j$ ⇒ $(2+2\varepsilon)$. Fully random h costs $m\log T$ bits.

KMV $h : [m] \to [m]$, $t = \frac{100}{\varepsilon^2}$, X = t-th smallest hash, output $\widetilde{DE} = \frac{tm}{X}$.
Under-estimate ⇔ fewer than t hashes below $\frac{tm}{(1-\varepsilon)DE}$; $Y = \sum Y_i$, $\mathbb{E}Y = \frac{t}{1-\varepsilon}$, $\mathrm{Var}Y \leq \mathbb{E}Y$ (pairwise) ⇒ Chebyshev $\leq \frac{1}{10}$; over-estimate same; union ⇒ 0.8; median ⇒ $1-\delta$. Space $O(\varepsilon^{-2}\log m \log\frac{1}{\delta})$. **!** fails under deletions

k-wise independent hash
$\Pr[h(x_1){=}y_1, \ldots, h(x_k){=}y_k] = m^{-k}$
for distinct x's. $h(x) = \sum_{j<k} c_j x^j$ over $\mathbb{F}_m$ (m prime). Proof: two polynomials agreeing on k points differ by a degree-$\leq k-1$ poly with k roots ⇒ equal; m^k polys ↔ m^k output tuples (bijection). Space $O(k\log m)$.

4 · L8 Communication complexity

Streaming → one-way CC Alice runs the algorithm on the first half, sends memory (S bits); Bob continues. CC lower bound C ⇒ space $\geq C$.

Equality det. $\geq n$: $\leq n-1$ bits ⇒ $\leq 2^{n-1}$ messages $< 2^n$ inputs ⇒ $x_1 \neq x_2$ same message; Bob with $y = x_1$ fails on one. Rand.: send $r \cdot x$ mod 2; $x \neq y$ caught w.p. $\frac{1}{2}$; log $\frac{1}{\delta}$ repeats; needs shared r.

Newman public → private with $+O(\log n + \log\frac{1}{\delta})$ bits, error $+\delta$: fix $t = 100n/\delta^2$ strings; Chernoff per input ($< 2^{-10n}$) + union over 2^{2n} inputs; Alice sends index i.

Hard problems Index (x / $i \to x_i$): $\Omega(n)$. Gap-Hamming (Hamming distance $\pm\sqrt{n}$): $\Omega(n)$.

Reduction A hard (C), extra message $|M|$ ⇒ B needs $C - |M|$.
Set intersection $\pm\frac{\sqrt{n}}{2}$: Ham $= |X|+|Y|-2|X\cap Y|$, send $|X| \Rightarrow \Omega(n)$.
Disjointness: $X = \{j : x_j = 1\}$, $Y = \{i\} \Rightarrow \Omega(n)$.

Graph streams Connectivity $\Omega(n)$: u_A-v_j iff $x_j = 1$ (Alice); u_B-v_j $\forall j \neq i$, u_A-u_B (Bob) ⇒ connected ⇔ $x_i = 1$. Perfect matching $\Omega(n^2)$: Index on n^2 bits as bipartite u_i-v_j; Bob attaches private dummies to all $u_{i\neq i^*}$, $v_{j\neq j^*}$ ⇒ PM ⇔ edge $u_{i^*}v_{j^*}$. MIS $\Omega(n^2)$ (two copies + special vertices).

5 · L9 Frequency estimation

Moments $F_p = \sum f_i^p$; F_0=distinct, F_1=length. Collision sampling needs $\Theta(\sqrt{n}/\varepsilon^2)$ — too many.

AMS (F_2) $p_i \in \{\pm1\}$, $Z = \sum p_i f_i$ (update $Z{+}{=}p_x\Delta$), output Z^2.
$\mathbb{E}Z^2 = \sum f_i^2$ (cross terms vanish); $\mathbb{E}Z^4 = \sum f_i^4 + 3\sum_{i\neq j} f_i^2 f_j^2 \leq 3F_2^2 \Rightarrow \mathrm{Var}Z^2 \leq 2F_2^2$.
Average $30/\varepsilon^2$ copies ⇒ Chebyshev $\leq \frac{1}{10}$. 4-wise signs. Space $O(\varepsilon^{-2}(\log m + \log n))$.

Models $\Delta = 1$ frequency; $\Delta \geq 0$ cash register; any turnstile; $x_i \geq 0$ always strict turnstile.

Linear sketch keep $y = Rx$; update y $+{=}$ $\Delta\, Re_i$ ⇒ deletions free. AMS ✓, KMV ✗. F_p ($p>2$): $\tilde{\Theta}(m^{1-2/p})$.

Misra-Gries $\leq k$ counters; present → $+1$; room → add; full → all -1, drop zeros. $\leq n/k$ decrement steps ⇒ $x_i - \frac{n}{k} \leq c_i \leq x_i$. $k = 1/\varepsilon$: error εn, deterministic, $O(\varepsilon^{-1}(\log m + \log n))$, insert-only, not linear. (Majority: one pair (i,c).)

Count-Min t rows $\times k$; $C_\ell[h_\ell(i)]$ $+{=}$ Δ; $\tilde{x}_i = \min_\ell$. Over-count mean $\leq \|x\|_1/k$; $k = 2/\varepsilon$ + Markov ⇒ row bad w.p. $< \frac{1}{2}$; $t = 2\log_2 m$ ⇒ $1/m^2$; union ⇒ $x_i \leq \tilde{x}_i \leq x_i + \varepsilon\|x\|_1$ $\forall i$. Strict turnstile only; pairwise hash.

Count Sketch also $g_\ell : [m] \to \{\pm1\}$; $C_\ell[h_\ell(i)]$ $+{=}$ $g_\ell(i)\Delta$; $Y_{\ell,i} = g_\ell(i)C_\ell[h_\ell(i)]$: $\mathbb{E} = x_i$, Var $\leq \|x\|_2^2/k$; $k = 10/\varepsilon^2$ ⇒ 0.9; median of $t = 100\log m$ ⇒ $|\tilde{x}_i - x_i| \leq \varepsilon\|x\|_2$ $\forall i$. Any turnstile.
ℓ_1 heavy hitters → Count-Min; ℓ_2 → Count Sketch.

Sparse recovery $\|x\|_0 \leq k$: lower bound $\Omega(k(\log m + \log n))$. 1-sparse: value $= \sum\Delta$; position bit b = [sum over indices with bit b set $>$ 0] (binary search). General: random 0/1 r separates $x \neq y$ w.p. $\geq \frac{1}{2}$; $O(k(\log m + \log n))$ fixed r's (exp. time).

6 · L10 Compression

Approx. sparse recovery $\mathrm{err}_2^k(x) = \min_{\|z\|_0 \leq k} \|z-x\|_2$ (keep top k). Count Sketch with $S = \frac{10k}{\varepsilon^2}$, $t = 100\log m$; keep top-k of x'.
Step 1: no collision with the k large items w.p. $\geq 1 - \frac{\varepsilon^2}{10}$; then Var $\leq \mathrm{err}^2/S \Rightarrow |x'_i - x_i| \leq \frac{\varepsilon}{\sqrt{k}}\mathrm{err}$ ($(k{+}2)$-wise).
Step 2: I = chosen, J = true top-k; $j \in J \setminus I$ displaced by $i \in I \setminus J$: $|x_j| \leq |x_i| + \frac{2\varepsilon}{\sqrt{k}}\mathrm{err}$ ⇒ total $\leq (1+9\varepsilon)\mathrm{err}^2 \Rightarrow (1+5\varepsilon)\mathrm{err}$. Space $O(\varepsilon^{-2}k\log m(\log n + \log m))$.

JL $\Pi_{ij} \sim \mathcal{N}(0,1)$, $k = \frac{\alpha}{\varepsilon^2}\log\frac{1}{\delta}$: $\|\frac{1}{\sqrt{k}}\Pi x\|_2 = (1\pm\varepsilon)\|x\|_2$ w.p. $1-\delta$ (independent of m). $y_i \sim \mathcal{N}(0, \|x\|_2^2)$; $\Pr[(1-\varepsilon)^2 k \leq \sum z_i^2 \leq (1+\varepsilon)^2 k] \geq 1 - 2e^{-c\varepsilon^2 k}$. n points: $\delta = \frac{1}{10n^2}$ + union ⇒ $k = O(\frac{\log n}{\varepsilon^2})$.

Regression $L(x) = \|Ax-b\|_2 = \|My\|_2$, $M = [A,b]$, $y = (x,-1)$. If $\|\Phi My\| = (1\pm\frac{\varepsilon}{10})\|My\|$ $\forall y$: $(1-\frac{\varepsilon}{10})L(\tilde{x}) \leq \tilde{L}(\tilde{x}) \leq \tilde{L}(x^*) \leq (1+\frac{\varepsilon}{10})L(x^*) \Rightarrow (1+\varepsilon)$.

Subspace embedding d-dim E, $k = O(\frac{d+\log(1/\delta)}{\varepsilon^2}) \Rightarrow$ all $x \in E$ preserved. Unit vectors only; $\frac{1}{2}$-net $|N| \leq 5^d$ (disjoint radius-$\frac{\gamma}{2}$ balls inside radius $1+\frac{\gamma}{2}$); JL on N and differences ($2^{O(d)}$ events) ⇒ inner products; $x = \sum z_i$, $\|z_i\| \leq 2^{1-i}$, $\sum\|z_i\| \leq 2 \Rightarrow$ error $O(\varepsilon)$. Sketch size $\tilde{O}(d^2/\varepsilon^2)$.

7 · Pitfalls

- Hash functions are fixed: repeated elements hit the same bucket (that is why hashing counts *distinct* elements).
- State the independence your analysis needs (pairwise / 4-wise / $(k+2)$-wise) and count its space.
- Count-Min needs $x_i \geq 0$; Misra-Gries is insert-only; KMV breaks under deletions.
- Morris: $\mathbb{E}[2^C] = A+1$, not A; output $2^C - 1$. Space is $\log\log n$ per copy, not $\log n$.
- Lower-bound reductions: check both directions (answer 1 $\Leftrightarrow$ $x_i = 1$) and that Alice's part depends only on x, Bob's only on i.
- Union bound over "all i" or "all inputs": per-item failure must be $\ll 1$/(number of items).

8 · Sketch comparison

Alg.	Model	Guarantee	Space
KMV	insert	$(1\pm\varepsilon)DE$	$\varepsilon^{-2}\log m$
AMS	turnstile	$(1\pm\varepsilon)F_2$	$\varepsilon^{-2}\log(mn)$
M-G	insert	$-\varepsilon n$, det.	$\varepsilon^{-1}\log(mn)$
CM	strict	$+\varepsilon\|x\|_1$	$\varepsilon^{-1}\log m\log(mn)$
CS	turnstile	$\pm\varepsilon\|x\|_2$	$\varepsilon^{-2}\log m\log(mn)$
JL	linear	$(1\pm\varepsilon)\|x\|_2$	$\varepsilon^{-2}\log\frac{1}{\delta}$ rows
Morris	insert	$(1\pm\varepsilon)A$	$\varepsilon^{-2}\log\log n$
Spanner	edge	stretch $2k-1$	$n^{1+1/k}\log n$

KMV/AMS: $\times\log\frac{1}{\delta}$ for $1-\delta$. M-G = Misra-Gries, CM = Count-Min, CS = Count Sketch.

9 · Answer templates

(A) Estimator analysis

1. Define indicator / signed variables; compute $\mathbb{E}$ (show unbiased or bound the bias).
2. Compute Var (expand square; cross terms vanish by pairwise / 4-wise independence).
3. Average k copies; Chebyshev $\Rightarrow$ constant success.
4. Median of $O(\log\frac{1}{\delta})$ $\Rightarrow 1-\delta$; union bound if needed for all items.
5. Space = copies × (counter bits + hash seed bits).

(B) Lower bound by reduction

1. Pick source: Index (one bit), Gap-Hamming (approximate count), Disjointness.
2. Alice turns x into stream part 1; Bob turns i into part 2.
3. Alice runs alg., sends memory; Bob finishes and reads answer.
4. Prove: output YES $\Leftrightarrow x_i = 1$ (both directions).
5. Conclude: space $\geq$ CC $= \Omega(\cdot)$; mention randomness $\Rightarrow$ randomized CC bound.

(C) Graph stream algorithm State what subgraph you keep and the rule for adding an edge; prove correctness via an invariant (forest / no short cycle / colouring); bound stored edges ($n-1$, or girth $> 2k \Rightarrow O(n^{1+1/k})$); space = edges $\times\, 2\log n$.

10 · Quick numbers

- $\frac{1}{e}\approx 0.368$, $\frac{1}{e^2}\approx 0.135$, midpoint ≈ 0.25
- Geometric guessing: $\log_{1+\varepsilon}\Delta = O(\frac{\log\Delta}{\varepsilon})$ thresholds.
- γ-net in d dims: $\leq (1+\frac{2}{\gamma})^d$ points.
- Storing a number $\leq N$: $O(\log N)$ bits; counters store $\log n$, ids store $\log m$.
- Hash k-wise over $\mathbb{F}_m$: k coefficients $\Rightarrow O(k\log m)$ bits.